# American Computer Science League

2021-2022 ● Contest 4: Short Problems ● Classroom Division

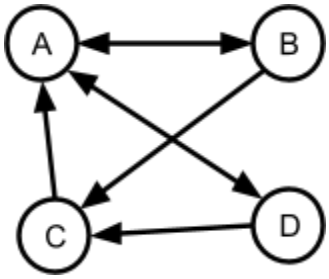
**1. Graph Theory**

Given vertices {A, B, C, D, E} and edges {AE, AB, BD, BA, CD, DC, BE, EC, CA}, how many paths of length 2 are there?

A. 12
B. 13
C. 14
D. 15
E. 16

**2. Graph Theory**

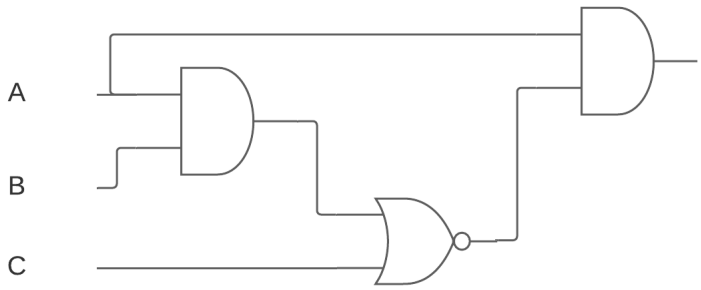
How many cycles are there from A in the directed graph below?




A. 2
B. 3
C. 4
D. 5
E. 6

**3. Digital Electronics**

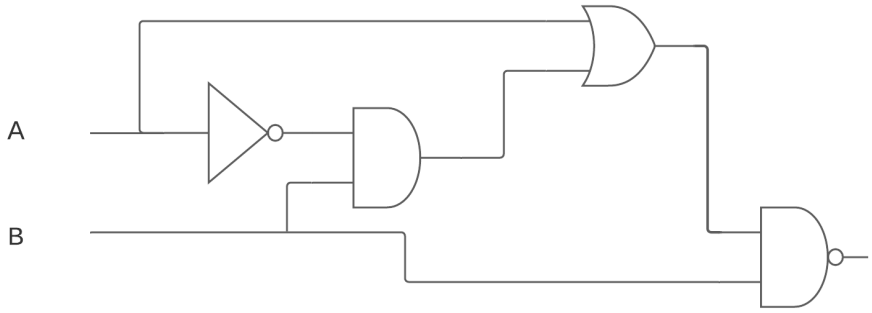
How many ordered triples make the Boolean expression represented by the following digital circuit TRUE?




A. 0
B. 1
C. 4
D. 5
E. 7

**4. Digital Electronics**

Simplify the Boolean expression represented by the following digital circuit.




A. $A + \overline{B}$
B. $\overline{AB}$
C. $\overline{A + B}$
D. $\overline{A}$
E. $\overline{B}$

**5. Assembly Language**

What is printed when the following program is executed?

```
N     DC      5
      LOAD    N
      STORE   F
      SUB     =1
TOP   STORE   B
      LOAD    F
      MULT    B
      STORE   F
      LOAD    B
      SUB     =1
      BE      OUT
      LOAD    B
      SUB     =1
      BU      TOP
OUT   PRINT   F
      END
```

A. 15
B. 30
C. 60
D. 120
E. 240

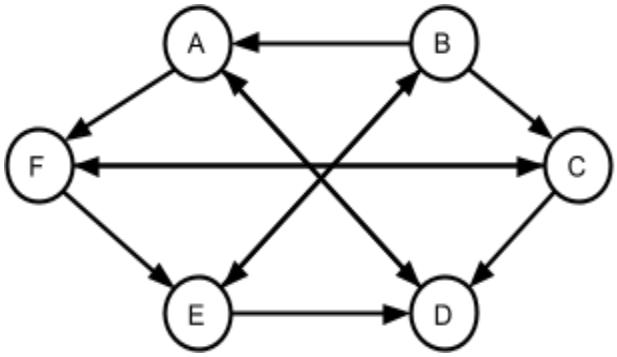
**6. Graph Theory**

Given a directed graph with vertices {A, B, C, D} and edges {AB, CB, AD, BC, BD, AC}, if edge DC were added, how many more paths of length 2 would there be?

A. 2
B. 3
C. 5
D. 6
E. 9

**7. Graph Theory**

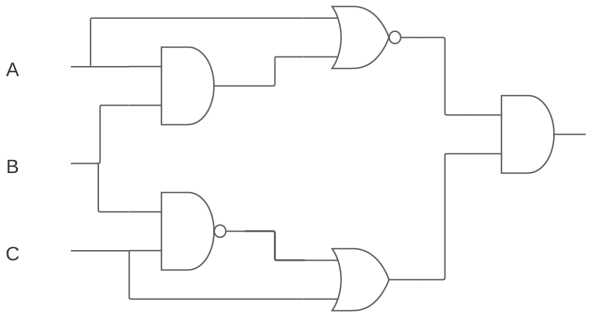
How many cycles of length 4 are there from A?




A. 1
B. 2
C. 3
D. 4
E. 5

**8. Digital Electronics**

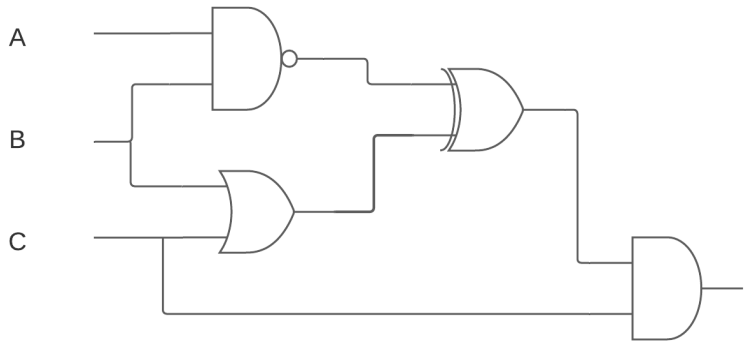
Simplify the Boolean expression represented by the following digital circuit.




A. $\overline{A}$
B. $\overline{A} + \overline{B}$
C. $\overline{B}$
D. $\overline{B} + \overline{C}$
E. $\overline{C}$

**9. Digital Electronics**

How many ordered triples make the following digital circuit TRUE?




A. 0
B. 1
C. 4
D. 7
E. 8

**10. Assembly Language**

How many values are printed when this program is executed?

```
A           DC          12
B           DC          1
TOP         LOAD        A
            DIV         B
            STORE       C
            LOAD        B
            MULT        C
            STORE       D
            LOAD        A
            SUB         D
            BE          OUT
IN          LOAD        A
            SUB         B
            BE          STOP
            LOAD        B
            ADD         =1
            STORE       B
            BU          TOP
 OUT        PRINT       B
            BU          IN
 STOP       END
```

A. 2
B. 3
C. 4
D. 5
E. 6